Daulet Imangaliyev

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Education

Politecnico di Milano — Milan, Italy
M.Sc. in Mathematical Engineering, Quantitative Finance Track — *Sep. 2026 – Present*

- Coursework (2026/27 study plan): Computational Statistics, Mathematical Finance II, Real and Functional Analysis, Stochastic Differential Equations, Advanced Programming for Scientific Computing (C++), Financial Engineering, Model Identification and Data Analysis.
- Student Associations: Polimi Data Scientists (PMDS) and Mathematical Engineers Association (AIM).

University of Bologna — Bologna, Italy
B.Sc. in Statistical Sciences, Stats&Maths Curriculum, 27.55/30, 110/110 — *Sep. 2023 – Oct. 2026*

- Relevant coursework: Statistics I–II, Mathematics I–II, Statistical Models, Multivariate Analysis, Econometrics, Time Series, Derivatives and Hedging, Informatics, Data Analysis, Advanced Analysis of Data.

University of Birmingham — Birmingham, United Kingdom
Erasmus Exchange Programme — *Sep. 2025 – Jun. 2026*

- Selected coursework: Mathematical Finance, Nonlinear Programming and Heuristic Optimisation, Applied Statistics, Statistical Methods in Economics, Numerical Methods and Numerical Linear Algebra.
- Student Societies: Member of Mathematics, Algorithms & Programming and Investment & Finance Societies.

Research Experience

Bachelor's Thesis — University of Bologna — 2026

A Controlled Evaluation of Static-Arbitrage-Free Option Surface Construction Methods
Supervisor: Prof. Antonio Castagna.

- Designed a controlled benchmark of three option-surface methods (constrained piecewise-linear, sequential eSSVI, SANOS), using **38.2 million** EOD SPX/SPXW option observations from OptionMetrics (2015–2024).
- Implemented leakage-controlled data preprocessing, held-out option-price evaluation, and independent static-arbitrage diagnostics.
- Evaluated 2,264 SPXW daily surfaces; SANOS achieved **1.30 normalized bps MAE**, with 92.0% of predictions inside quoted bid–ask intervals in the reported benchmark.

Nazarbayev University — Institute of Smart Systems and AI (ISSAI) — Astana, Kazakhstan
Undergraduate Research Assistant — *Jun. 2023 – Jun. 2024*

- Fine-tuned Meta MMS and OpenAI Whisper for 22 languages, **reducing WER and CER by 5–15%**.
- Parallelized multi-GPU training and **reduced Docker-container RAM usage by 50%**.
- Built a web crawler collecting 1+ TB of Kazakh text from 200,000+ URLs, **increasing throughput by 40%**.

Selected Projects

Yield Spread Time-Series Analysis | R, forecast, ggplot2, dplyr — Feb. – May 2024

- Built a pipeline to analyse 50+ years of U.S. 10Y–2Y Treasury yield-spread data from FRED.
- Applied ADF/KPSS stationarity tests, seasonal and spectral decomposition, and ARMA/ARIMA forecasting; compared model specifications using BIC and RMSE.

Professional & Academic Engagement

Lead The Future Mentorship | Selected Mentee — 2026 – Present
Selected for a competitive one-to-one STEM mentorship programme (**13% historical acceptance rate**).

XIII World Congress of the Bachelier Finance Society | Attendee — Jun. – Jul. 2026
International research congress covering stochastic modelling, volatility, market microstructure, and quantitative finance.

ALGODEFI26, Politecnico di Milano | Attendee — Oct. 2026
Workshop on algorithmic trading, automated market making, machine learning, and digital assets.

Technical Skills

Programming & Data: Python (NumPy, pandas, Polars), R, SQL, C++, MATLAB.

Statistical & Numerical Methods: Statistical Modelling, Time-Series Analysis, Econometrics, Numerical Optimisation, Forecasting, Model Validation.

Machine Learning: PyTorch, TensorFlow, XGBoost, LightGBM, Model Training & Evaluation.

Quantitative Finance: Derivatives Pricing, Implied Volatility Modelling, Option Surface Calibration.

Tools: Git, Linux, Docker.